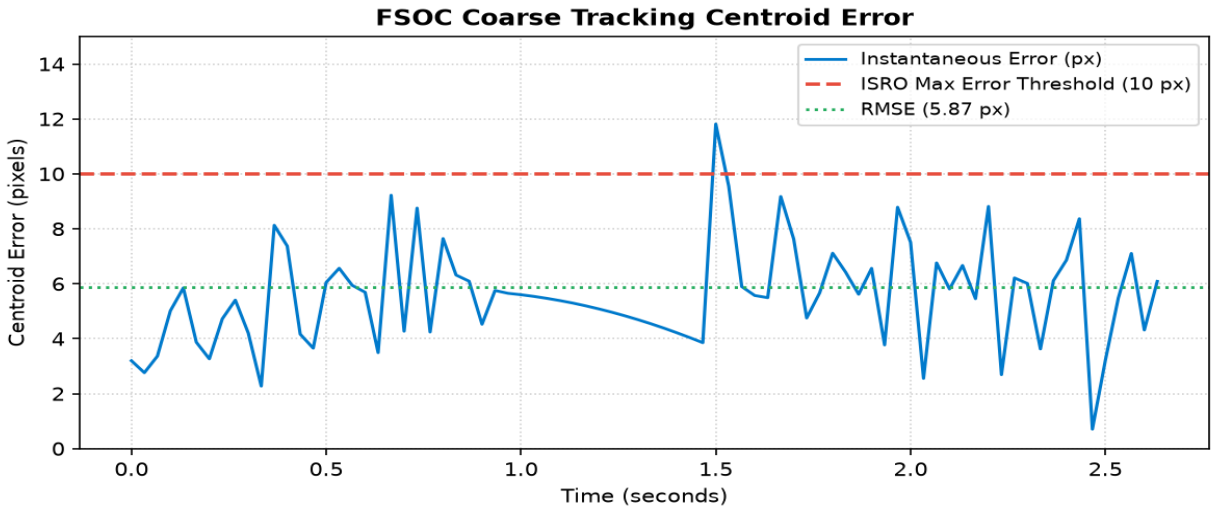
ISRO Smart India Hackathon (SIH) Evaluation Report

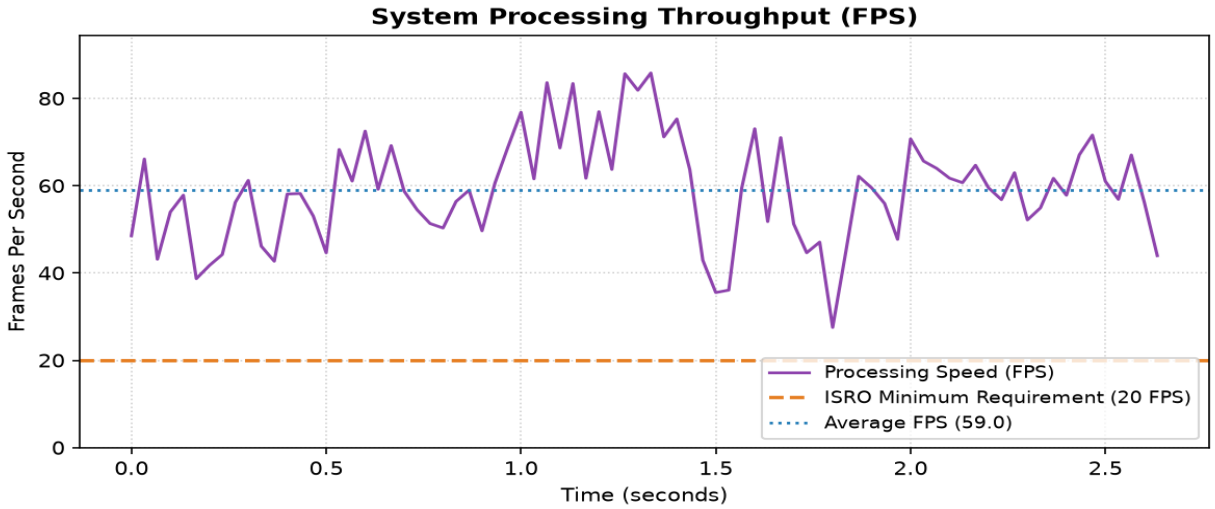
AI-Based Virtual Camera Tracking System for Coarse Alignment of Mobile FSOC Terminals

Evaluation Parameter	ISRO SIH Specification	Measured Value	Status
Tracking Centroid Error (RMSE)	≤ 10.0 pixels	5.87 px	PASS
Processing Speed	≥ 20.0 FPS	59.0 FPS	PASS
Lock Retention Rate	> 95.0%	98.8%	PASS
Acquisition Time	≤ 2.0 seconds	0.088 s	PASS
Re-acquisition Time	≤ 1.0 second	0.035 s	PASS

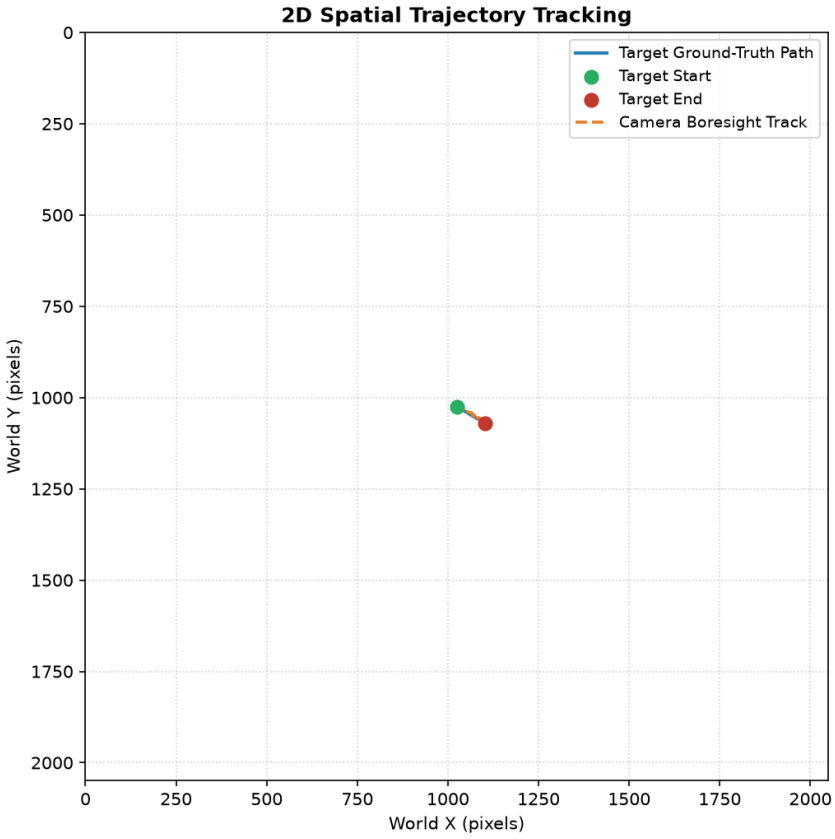
Tracking Error Analysis:



Processing Throughput (FPS):



2D Spatial Trajectory Ground Truth vs Camera Boresight:


2D Spatial Trajectory Tracking
Target Ground-Truth Path
Target Start
Target End
Camera Boresight Track
World Y (pixels)
World X (pixels)
0
250
500
750
1000
1250
1500
1750
2000